

SWOT analysis for the Indian PCB industry

	Helpful	Harmful
Internal	Strengths ➤ Flexibility in terms of handling high-mix, low-volume batches ➤ Ability to face frequent 'pull and push' from customers ➤ Readiness to adopt design changes ➤ Flexible payment terms ➤ Infrastructure to effectively handle the production of low- and mid-range PCBs in low and medium quantities ➤ High quality of products, unlike the PCBs imported from China ➤ Good customer services and technical support ➤ Ability to customise products and offer specialised processes for the customer	Weaknesses ➤ Lack of automation and technological upgrades ➤ Inability to produce mass volumes ➤ Inherent absence of infrastructure and ecosystem for creating PCBs in large quantities and of high complexities ➤ Insufficient go-to-market expertise ➤ Lack of educated, skilled and trained manpower ➤ Lack of strong investments and affordable finance ➤ High cost of production ➤ Low export incentives compared to competitors like China
	Opportunities ➤ Increasing demand for PCBs in India ➤ Popularity for domestically manufactured PCBs in low-end and mid-range applications ➤ Import not feasible for low- technology boards in small batches ➤ Positive and responsive approach of Department of Electronics and Information Technology (DeitY) ➤ Modified Special Incentive Package Scheme (M-SIPS) by government of India ➤ Value consciousness of customers ➤ Requests for highly customised, high-quality boards for specific applications	Threats ➤ Taxation and excise issues with raw materials ➤ Dependency on import of raw materials, and production and testing equipment ➤ High cost and insufficient availability of power ➤ Environmental measures and lag in implementing them ➤ Duty-free PCB imports being dumped in the market ➤ Bureaucratic hurdle: lengthy and time-consuming processes for approvals and clearances of new projects

sion makers believe that they have been instrumental enough. DeitY has set electronic hardware production target of US\$ 400 billion by year 2020 which, if it materialises, will create huge demand for PCBs. "The department is positive, responsive and very keen in promoting electronic manufacturing in India," says Satish Athavale, managing director, Shogini Technoarts Pvt Ltd.

Indian Printed Circuit Association (IPCA) is a non-profit, voluntary, professional organisation formed in 1985 with a mission to promote the PCB industry in India and represent their interests. The members of this organisation include PCB manufacturers, designers, electronic manufacturing services (EMS) companies and suppliers of PCB machines and raw materials.

What's holding India PCB Inc. back?

One of the biggest drawbacks is that the Indian industries inherently lack

the ecosystem or infrastructure for large-scale and most advanced manufacturing techniques. Most importantly, the lack of investment incentives to bear the risk in India hampers growth of the PCB industry here.

Indian PCB manufacturers can, generally, handle small and medium batches effectively. They are flexible enough to accommodate design changes at lower tool cost and can work with smaller panel sizes to produce small to mid-size quantities. However, high-end and large batches of PCBs are still imported in most cases.

Automation and adaption of the latest technologies is slow in the industry. One of the reasons is the lack of good investment. Though PCB industry has lot of potential, convincing potential investors has been a challenge.

The electronics manufacturing in the country drives demand for PCBs. As manufacturing has remained low the demand for PCBs has also remained low.

Another issue is the changes and

Which industry shows the highest PCB demand?

Most of the PCB demand in India comes from mobile phones, consumer electronics, automotive, power electronics and industrial automation segments. Energy meters and telecommunication industry come next. It is expected that in the next five years, demand for PCBs would considerably increase from LED lighting and solar sectors. Currently two million mobile phones are being imported into India. Indian PCB manufacturers are gearing up to this demand while some claim to already have the capability to meet the demand.

Reasons for low manufacturing base in India and lack of competitiveness

- Large capital investments required for setting up PCB manufacturing units is inhibiting manufacturers to set up new units, especially for multilayered PCBs, whose demand is expected to grow at a significant rate.
- High investment required in pollution control equipment due to stringent pollution standards and polluting nature of industry.
- Inverted duty structure, as inputs for manufacture of PCBs are subject to 5% to 10% customs duty while PCBs attract nil duty under the ITA-1.
- Lack of availability of skilled labour.
- Lack of availability of raw materials like pure-grade silicon and copper laminates required for fabrication of the printed circuit boards.
- Lack of availability of raw materials at competitive prices.
- Lack of access to new technology and state-of-the-art capital equipment.

— Rajoo Goel, secretary general, ELCINA Electronic Industries Association of India